HYDROFARMS AGRI SOLUTIONS

EVAPORATIVE COOLING, EXHAUST VENTILATION AND AIR CIRCULATION

Activity-Specific Construction Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-CLM-001
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt - open terrain
Outdoor design basis	40 degC dry-bulb; actual wet-bulb recorded during performance test
Prepared by	Hydrofarms Agri Solutions
Issue date	15 August 2026
Parent methods	HF-GH01-MST-STR-001, CIV-003 and ENV-001

PERFORMANCE IS MEASURED AT INSTALLED RESISTANCE
Free-air fan capacity is not an acceptance value. Fan duty shall be verified with the installed shutters, guards, wetted pads, insect net and envelope leakage at the measured system static pressure.

Document Control

Approval and responsibility

Role	Primary responsibility
Project manager	Resources, procurement status, subcontractor control and programme.
Construction manager	Work-front release, sequencing, access and trade coordination.
Mechanical engineer	Equipment selection, hydraulic/airflow checks, installation and commissioning.
Structural engineer	Fan/pad chassis, anchors, reactions, vibration and opening-frame release.
Electrical/controls engineer	Starters, protection, isolators, sensors, interlocks and alarms.
Water-quality specialist	Make-up water suitability, EC-based bleed and scale control.
QA/QC engineer	Material traceability, inspections, tests, balancing and turnover records.
HSE manager	Lifting, rotating equipment, electrical isolation, chambers and work at height.
Client/consultant	Hold/witness inspection and performance acceptance.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method defines supply-interface checks, installation, alignment, hydraulic testing, electrical/controls coordination, balancing and performance acceptance of the production and service greenhouse exhaust fans, internal circulation fans, evaporative cooling pads, water-distribution/return systems, pad tanks and pumps. The objective is controlled negative-pressure airflow through wetted pads without uncontrolled bypass, dry pad zones, water carryover or unsafe equipment operation.

2. Scope and Package Boundary

- Thirty-five production and three service-greenhouse exhaust fans, shutters, guards, vibration isolation, dedicated ground-supported chassis interfaces and perimeter seals.
- Twenty-eight production air-circulation fans and two service-greenhouse circulation fans, including guards and galvanized suspension hardware.
- Production and service cooling-pad media, 30 mm distribution media, treated-sheet housings, perforated top headers, lower collection gutters and hydraulic fittings.
- Four independent production pad-water circuits and one independent service circuit, including operating tanks, duty pumps, warehouse spares, filters, valves, instruments, make-up, bleed and drain interfaces.

DUPLICATION CONTROL
Concrete tank chambers, waterproofing, sumps and embedded civil items are measured in civil works. Fan/pad support steel is measured in structural works. Xsect AirCool material is measured in covering works. Motor feeders, final cables, starters/panels, sensors and central automation are measured in electrical/controls. This method coordinates and commissions these interfaces without pricing them twice.

3. Governing Information and Technical Basis

Priority	Controlled information
1	Applicable Egyptian statutory requirements and approved AFC drawings
2	Certified supplier fan curves, pad data, pump curves and approved equipment schedule
3	Approved structural opening/chassis drawings and pad housing/hydraulic shop drawings
4	Approved electrical/control single lines, motor schedule and cause-and-effect matrix
5	ANSI/AMCA 210-25 / ASHRAE 51-25 for certified aerodynamic fan performance
6	AMCA Publication 211-26 certified-ratings programme requirements where specified
7	Manufacturer installation, operation, water-quality and maintenance instructions

DESIGN STATUS
The adopted equipment quantities exceed the original theoretical 21-fan review. The owner-approved 35-fan production arrangement is retained for staged operation, distribution and redundancy. Final fan model approval remains conditional on certified duty at actual installed static pressure.

4. Adopted System Geometry

Parameter	Production greenhouse	Service greenhouse
Footprint	126 m x 36 m	9 m x 36 m on west side
Eaves / crown	5.50 m / 7.00 m	5.50 m / coordinated roof crown
Air path	North pad wall to south fan wall; 36 m travel	Pad side to three-fan wall
Pad face	Approx. 125 m x 3.0 m x 0.15 m	9 m x 3.0 m x 0.15 m
Pad maintenance aisle	0.75 m clear	Coordinated clear service access
Fan-side aisle	0.75 m clear	Coordinated clear service access

5. Equipment and Performance Schedule

Code	Equipment	Quantity	Adopted performance / arrangement
CLM-01	Production exhaust fans	35	54 in / 1.40 m class; 21,300-24,800 CFM each at installed duty
CLM-02	Service exhaust fans	3	Same approved duty class; triangular three-fan arrangement
CLM-03	Production circulation fans	28	800 mm; 8,000-10,000 m3/h; 0.37-0.55 kW
CLM-04	Service circulation fans	2	600 mm; 4,000-6,000 m3/h; 0.25-0.37 kW
CLM-05	Production pad media	375 m2	150 mm, 7090-type; 418 installed 600 x 1,500 mm modules
CLM-06	Service pad media	27 m2	150 mm, 7090-type; 30 installed 600 x 1,500 mm modules
CLM-07	Distribution-pad strips	224	30 mm; 209 production + 15 service modules
CLM-08	Production pad tanks	4 x 10 m3	One LLDPE/HDPE tank per hydraulically independent sector
CLM-09	Production pad pumps	4 duty + 1 spare	2.2 kW; 18-20 m3/h at 10 m; about 3 HP
CLM-10	Service pad tank/pumps	1 m3; 1 duty + 1 spare	0.75 kW; 5-6 m3/h at 10 m; about 1 HP

5.1 Production airflow check

Calculation	Result
Minimum total airflow	35 x 21,300 = 745,500 CFM = approximately 1,266,000 m3/h
Maximum total airflow	35 x 24,800 = 868,000 CFM = approximately 1,475,000 m3/h
Approximate greenhouse volume	29,500 m3 including arched-roof allowance
Air changes	Approximately 42.9-50.0 ACH
Air replacement time	Approximately 1.40-1.20 minutes
Production pad face velocity	Approximately 0.94-1.09 m/s across 375 m2
Service pad face velocity	Approximately 1.12-1.30 m/s across 27 m2

These calculations demonstrate installed air capacity; they do not guarantee crop-level temperature. Actual pad-leaving temperature and the rise across the 36 m crop path depend on outdoor wet-bulb temperature, solar load, screen position, crop density, pad cleanliness, leakage and measured resistance.

6. Fan Layout and Mechanical Installation

6.1 Exhaust fan wall

- Arrange production fans in alternating 2-fan and 3-fan bays over fourteen 9 m bays, giving 7 x 2 + 7 x 3 = 35 fans. Group operation as 9 + 9 + 8 + 9 fans for staged control.
- Set lower fan centrelines at 2.40 m AFFL. In each 3-fan bay set the upper fan centreline at approximately 4.40 m AFFL, subject to the approved chassis/fan shop drawing.
- Install the three service fans in the approved triangular arrangement using the same lower/upper centreline logic and an independent service control sequence of 1 + 2.
- Each fan chassis shall extend to and anchor at the floor/foundation. Fan weight and vibration shall not be carried by greenhouse plastic, wall rails or light cladding members.
- Level and square the casing; fit vibration isolators, EPDM/PU perimeter seals, freely moving gravity shutters, internal/external guards and accessible belt/motor service panels.
- Verify blade clearance, rotation, belt alignment/tension where applicable, motor current and shutter opening before continuous operation.

6.2 Internal air-circulation fans

- Install two 800 mm fans per 9 m production bay, giving 28 fans, generally at 3.20-3.50 m centreline and directed from the pad side toward the exhaust side.
- Locate one 600 mm service fan in the germination/nursery portion and one in the packing portion without producing damaging direct jets at plants or workers.
- Use corrosion-resistant guarded fans with galvanized chains, hooks, safety cables and adjustable brackets fixed only to approved structural points.
- Align air streams to overlap and remove stagnant zones without short-circuiting directly into exhaust fans. Confirm final direction by smoke and velocity mapping.

7. Cooling-Pad Media and Housing

Component	Adopted requirement
Main pad media	600 x 1,500 x 150 mm cellulose modules; two rows high for 3.0 m production/service walls
Installed modules	418 production + 30 service; procure approved maintenance spares separately
Distribution media	30 mm thick, same flute/section; 209 production + 15 service modules
Housing/gutters	Minimum 2.0 mm Z275 galvanized treated sheet; minimum 120 micron dry epoxy coating
Fasteners/seals	SS304 fasteners and continuous EPDM-compatible water seals
Lower gutter	Minimum clear 200 x 200 mm; 0.5% fall to sector return outlets
External protection	Xsect AirCool enclosure and maintenance access coordinated with ENV-001

- Check flute orientation and airflow arrow on every module. Install rows continuously with tight vertical joints but without crushing, shaving or deforming the media.
- Provide uniform top and bottom clamping, removable service sections and separation between treated sheet, pad media and incompatible metals.
- Fit the 30 mm distributor media directly below the perforated header to spread water before it enters the main pad face.
- Complete a dry visual check, then a wetting test. No water may spray directly into the greenhouse, bypass the media, leak from joints or remain trapped in the housing.

8. Pad-Water Circuits

8.1 Hydraulic arrangement

Service	Production sectors	Service greenhouse
Number of independent circuits	4	1
Tank	10 m3 LLDPE/HDPE each	1 m3 LLDPE/HDPE
Pump	2.2 kW; 18-20 m3/h at 10 m	0.75 kW; 5-6 m3/h at 10 m
Pump discharge	2.5 in uPVC PN10	1.5 in uPVC PN10
Top distribution header	2 in perforated uPVC	1.25 in perforated uPVC
Gravity return	4 in uPVC	3 in uPVC
Drain/flush	1.5 in	1 in
Perforated-header length	125 m total	9 m

- Centre-feed each perforated header. Initial perforation is 3 mm at 100 mm centres, then field-balance flow to obtain an unbroken wetting line without direct jetting.
- Support all pipework independently of pad media and gutters. Provide full-bore isolation, unions/dismantling joints, silent check valve on discharge, pressure gauge, flow indication and sample point.
- Provide a manual duplex 120-mesh disc-filter arrangement per operating circuit with isolation and a flushing bypass so one element can be serviced without contaminating the header.
- Maintain continuous gravity fall from each 200 x 200 mm collection gutter to its correct tank. No cross-connection is permitted between sectors or between production and service circuits.
- Provide a 2 in make-up interface rated 15 m3/h at 2-3 bar, with 1.5 in production branches and 1 in service branch. Make-up treatment ownership remains under the water-quality chapter.
- Provide manual EC-based bleed through a 2 in discharge to the lined evaporation basin. No pad bleed, wash water or tank overflow may enter nutrient or reserve-water tanks.

8.2 Underground tank-room interfaces

Each production pad tank is installed in a separate open-top underground room, preliminary clear size 4.00 x 4.00 x 3.50 m. The service tank room is preliminarily 2.50 x 2.50 x 2.50 m. Final dimensions follow selected tank and pump-removal clearances. Civil construction and waterproofing remain under CIV-003.

- Confirm tank support is level and continuous, chamber floor falls to the sump, safe access/ladder and ventilation are complete before mechanical installation.
- Provide one float-operated sump pump for each approved room/compartment, a 2 in discharge with check valve/isolation/union, IP65 lighting and a maintenance chain block or portable lifting arrangement.
- Keep pump suction flooded and short; prevent air pockets with correct reducer orientation and provide low-level dry-run protection.

9. Controls and Staged Operating Sequence

Stage	Function	Operating logic
S0	Minimum circulation	Circulation fans operate; temperature/RH logging active
S1	Dry ventilation	First exhaust-fan group starts; pads remain dry
S2	Cooling enable	Required pad pumps start; allow 30-60 s pre-wet before adding fans
S3	Medium load	Stage additional fan groups while monitoring static pressure and RH
S4	Maximum load	All required fan groups and pad sectors operate
Alarm	Abnormal condition	High temperature, low water, pump trip, phase failure or filter alarm

- Provide four production temperature/RH points distributed along the airflow path, one shielded outdoor point and one service-greenhouse point.
- Provide low/high level switches for every pad tank and differential-pressure sensing or a manometer across the pad wall.
- Every fan and pump shall have lockable local isolation and Manual-Off-Auto control. PLC/controller failure shall not prevent authorized manual ventilation.
- Pad pumps shall be interlocked to minimum exhaust airflow; low tank level stops the associated pump but shall not stop emergency dry ventilation.
- Electrical panels, feeders, final cables, alarms and GSM/event logging are delivered under electrical/automation, with this method defining the functional cause-and-effect test.

10. Pre-Commissioning Inspection

- Confirm all fan/pad frames, anchors, coating repairs, guards, shutters, flexible seals, pipe supports and access clearances are complete.
- Flush tanks, filters, pipes, headers, gutters and returns. Remove swarf, coating debris, sealant and construction dust before pad wetting.
- Pressure-test pump discharge pipework to the approved test pressure; separately water-test gravity returns, gutters and tank connections.
- Check motor insulation, phase sequence, overload settings, earthing, local isolators, controls and rotation with drives safely isolated from personnel.
- Fill each circuit independently, vent pumps and lines, prove low/high level switches, then verify uniform make-up and controlled bleed paths.

11. Performance Testing and Acceptance

No.	Test	Acceptance criterion
01	Mechanical run	Correct rotation; free shutters; stable current; no abnormal noise, belt slip or vibration
02	Pad wetting	Uniform wetting within 3-5 min; no dry zone, direct spray, carryover or gutter leak
03	Airflow/static pressure	Each fan duty reconciles with certified curve at measured installed pressure
04	Temperature/RH traverse	Record outdoor, pad-leaving, mid-path and fan-end dry-bulb, RH and wet-bulb
05	Velocity mapping	Grid across width/height; target local deviation within +/-15% of measured mean
06	Control/interlock test	All stages, low level, trips, phase loss, manual mode and alarms demonstrated
07	Eight-hour run	Summer-duty operation without overheating, overflow, leakage, blocked filter or drift

NO UNCONTROLLED AIR BYPASS
During peak-performance testing, doors and envelope openings shall be in their normal sealed operating condition. Replacement air shall enter through the wetted pad systems; bypass leakage shall be located and rectified before final acceptance.

12. Inspection and Test Plan

No.	Inspection stage	Method	Type	Record
01	Equipment data, fan/pump curves and shop drawings	Document review	H	Material/design release
02	Structural chassis/openings and civil rooms	Survey/visual	H	Interface release
03	Delivered fans, pads, tanks, pumps and pipework	Identity/condition	W	Delivery inspection
04	Fan alignment, guards, shutters and vibration isolation	Visual/mechanical	H	Fan release
05	Pad housing, media, distributor and gutter	Dimensional/visual	H	Pad-wall release
06	Pressure and gravity water circuits	Pressure/water test	H	Hydraulic test report
07	Electrical protection, rotation and local controls	Electrical/functional	W	Pre-start checklist
08	Wetting/balancing and water carryover	Operational/visual	H	Wetting report
09	Airflow, static pressure and climate traverse	Instrumented test	H	Performance report
10	Eight-hour run, training and handover	Operational/records	R	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

13. Hold Points

- HP-CLM-01: Certified performance data, coordinated shop drawings and equipment selections accepted before procurement.
- HP-CLM-02: Fan chassis/openings, pad frames and tank rooms accepted before equipment installation.
- HP-CLM-03: Fan guards, shutters, rotation and mechanical alignment accepted before continuous operation.
- HP-CLM-04: Pad media, distributor, housing, gutter and net/envelope interfaces accepted before wetting.
- HP-CLM-05: Pressure/gravity tests, flushing and tank-level protection accepted before system operation.
- HP-CLM-06: Airflow/static-pressure, climate traverse and eight-hour tests accepted before handover.

14. HSE and Environmental Controls

Hazard	Mandatory control	Residual risk
Rotating fans	Guards, isolation/LOTO, proven stop, controlled commissioning zone	Medium
Lifting fans/pads/pumps	Lift plan, certified rigging, tag lines, stable access and exclusion zone	Medium
Underground rooms	Safe edge/access, ventilation, rescue plan, dry floor and protected opening	Medium
Electrical/wet systems	IP-rated equipment, earthing, RCD where required, dry test practice and LOTO	Medium
Sharp sheet/media	Cut-resistant gloves, deburring, protected edges and controlled handling	Low
Water treatment/biocide	Approved SDS, dosing control, PPE and disposal to approved route	Medium
Noise	Measured exposure, hearing protection and restricted access during full-load test	Low
Bleed/wash water	Route only to lined evaporation basin; prevent soil and nutrient-system discharge	Low

15. Records, Spares and Handover

- Approved fan/pump curves, pad data, equipment schedules, shop drawings, hydraulic calculations and cause-and-effect matrix.
- Delivery records, motor/pump nameplates, material certificates, coating repairs and installation inspections.
- Pressure, gravity-return, gutter, wetting, motor-current, vibration, airflow, static-pressure and climate-traverse reports.
- Final balancing settings, perforation corrections, valve positions, filter data, level/interlock settings and alarm tests.
- Eight-hour run report, as-built drawings, O&M manuals, warranties and operator training records.
- Stored spares: one matching production pad pump, one matching service pad pump, approved spare pad modules, belts/bearings, filter elements, level switches, seals and critical electrical spares.